

Reviewed Preprint

v1 • December 30, 2025

Not revised

Reviewed Preprint

v2 • April 16, 2026

Revised by authors

Prolonged oscillating preoptic area kisspeptin neuron activity underlies the preovulatory luteinizing hormone surge in mice

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Competing interests: No

competing interests declared

Funding: See [page 20](#)

Reviewing editor: John Ewer,
Universidad de Valparaíso, Chile

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eLife Assessment

This **fundamental** work advances our understanding of the role of kisspeptin neurons in regulating the luteinizing hormone (LH) surge in females. The study uses cutting-edge techniques to provide **compelling** and rigorous data supporting a critical role of RP3V kisspeptin neurons in the neuroendocrine LH surge process. This research will be of interest to reproductive biologists and neuroscientists studying the female ovarian cycle. Continuing to examine the complexities of the LH surge and the neuronal populations involved, as done in this study, is critical for developing therapeutic treatments for women's reproductive disorders.

<https://doi.org/10.7554/eLife.109215.2.sa2>

Abstract

The population of kisspeptin neurons located in the rostral periventricular area of the third ventricle (RP3V) is thought to have a key role in generating the GnRH surge that triggers ovulation. Using a modified GCaMP fibre photometry procedure, we have been able to record the in vivo population activity of RP3V^{KISS} neurons across the estrous cycle of female mice. A marked increase in GCaMP activity was detected beginning on the afternoon of proestrus that lasted in total for 13±1 hours. This was comprised of slow baseline oscillations with a period of 91±4 min and associated with high frequency rapid transients. Very little oscillating baseline or transient activity was detected at other stages of the estrous cycle. Concurrent blood sampling showed that the peak of the LH surge occurred 3.5±1.1 h after the first baseline RP3V^{KISS} neuron baseline oscillation on the afternoon of proestrus. The time of onset of RP3V^{KISS} neuron oscillations varied between mice and across subsequent proestrous stages in the same mice. To assess the impact of estradiol on RP3V^{KISS} neuron activity, mice were ovariectomized and given an incremental estradiol replacement regimen. Minimal patterned GCaMP activity was found in OVX mice, and this was not changed acutely by any of the estradiol treatments. However, on the afternoon of the expected LH surge, the same oscillating baseline activity with associated transients occurred for 7.1±0.5 h. These observations reveal an unexpected prolonged oscillatory pattern of RP3V^{KISS} neuron activity that is dependent on estrogen and underlies the preovulatory LH surge as well as potentially other facets of reproductive behavior.

Introduction

A neural surge generator is responsible for integrating multiple signals to control the mid-cycle activation of GnRH neurons that triggers the preovulatory luteinizing hormone (LH) surge (Herbison 2015 [↗](#), Kauffman 2022 [↗](#)). It now seems clear that a preoptic area population of kisspeptin neurons that directly activate GnRH neuron cell bodies (Piet, Kalil et al. 2018 [↗](#)) is an essential component of the surge generator in spontaneously ovulating mammals (Matsuda, Ohkura et al. 2019 [↗](#), Goodman, Herbison et al. 2022 [↗](#)).

The primary regulator of surge generator activity in spontaneously ovulating species is the gradually increasing levels of circulating estradiol that occur throughout the follicular phase of the cycle. In rodents, this “estrogen positive feedback” mechanism is mediated by estrogen receptor alpha (ESR1) and occurs within the rostral periventricular area of the third ventricle (RP3V) (Wintermantel, Campbell et al. 2006 [↗](#), Porteous and Herbison 2019 [↗](#)); a preoptic brain region encompassing the anteroventral periventricular nucleus (AVPV) and periventricular preoptic nucleus (PeN) (Herbison 2008 [↗](#)). In vivo CRISPR/Cas9 gene editing to knockdown ESR1 expression selectively in RP3V kisspeptin (RP3V^{KISS}) neurons has been shown to suppress both the LH surge and estrous cyclicity in adult female mice (Wang, Vanacker et al. 2019 [↗](#), Clarkson, Yip et al. 2023 [↗](#)). Although of variable importance amongst mammalian species, a circadian input to the surge generator is also critical for the GnRH surge in rodents (Goodman, Herbison et al. 2022 [↗](#)). It remains unclear how this operates, but a vasopressin input from the suprachiasmatic nucleus to the RP3V^{KISS} neurons is suspected to be important in triggering the onset of the surge (Tonsfeldt, Mellon et al. 2022 [↗](#), Piet 2023 [↗](#)).

The ability to monitor the activity of arcuate nucleus (ARN) kisspeptin neurons in freely behaving mice with GCaMP fibre photometry has been invaluable in defining and then understanding their role as the GnRH pulse generator in puberty and adulthood, as well as in pathological states (Clarkson, Han et al. 2017 [↗](#), Han, Kane et al. 2019 [↗](#), McQuillan, Han et al. 2019 [↗](#), Liu, Yeo et al. 2021 [↗](#), McQuillan, Clarkson et al. 2022 [↗](#), Goto, Hagihara et al. 2023 [↗](#), Han, Morris et al. 2023 [↗](#), Goto, Hagihara et al. 2025 [↗](#), Zhou, Han et al. 2025 [↗](#)). To date, attempts to record the behaviour of the RP3V^{KISS} neurons have been unsuccessful due primarily to the vertical column-like topography of the RP3V^{KISS} neurons alongside the third ventricle. Indeed, attempts by us to employ bevelled or mirror lenses that gather fluorescence from the side of the optic fibre have been met with very limited success. We now report here the use of tapered optic fibres (Pisano, Pisanello et al. 2019 [↗](#)) that enable the population activity of RP3V^{KISS} neurons to be monitored in real time in freely behaving mice. This reveals the estrogen-dependent dynamics of RP3V^{KISS} neuron population activity across the estrous cycle and reveals an unexpected oscillatory pattern of synchronised activity that continues for over 12 hours on proestrus.

Results

Characterization of GCaMP expression in RP3V^{KISS} neurons

Adult *Kiss1*^{Cre/+} (Palmiter v2) female mice (Padilla, Johnson et al. 2018 [↗](#)) were injected bilaterally with recombinant Cre-dependent AAVs encoding GCaMP6s. As the relationship of Cre-driven expression to kisspeptin immunoreactivity has not been reported for the RP3V in this line, we undertook dual label immunofluorescence for GFP (GCaMP6s) and kisspeptin in 5 female mice killed on proestrus. All mice exhibited many GFP-immunoreactive cells (33.0 ± 5.9 cells/section) located primarily adjacent to the third ventricle within the preoptic area (Fig.1A [↗](#)) and this exhibited overlap with kisspeptin immunoreactivity (21.8 ± 3.1 cells/section) (Fig.1B&C [↗](#)). However, the numbers of clearly identifiable dual labelled cells were 10.4 ± 2.5 /section resulting in $47.0 \pm 5.7\%$ of kisspeptin-immunoreactive neurons expressing GCaMP6s and $33.0 \pm 5.5\%$ of GFP cells expressing clearly cytoplasmic kisspeptin immunoreactivity. These relatively low levels of Cre-targeted expression, compared to ARN kisspeptin neurons, are typical for RP3V kisspeptin neurons in other *Kiss1*-Cre lines (Yip, Boehm et al. 2015 [↗](#), Yeo, Kyle et al. 2016 [↗](#), Piet, Kalil et al.

2018). This presumably results from the difficulty of detecting kisspeptin immunoreactivity in the cytoplasm of all RP3V kisspeptin neurons. *Kiss1-Cre* negative proestrus female mice given AAV injections (N=2) did not exhibit any GFP/GCaMP6s expression.

RP3V^{KISS} neuron population activity across the estrous cycle

To make fibre photometry recordings from RP3V kisspeptin neurons, mice were given unilateral injections of AAV9-CAG.FLEX.GCaMP6s into the RP3V of adult *Kiss1^{Cre/+}* female mice followed by the implantation of a tapered optic fibre (Fig.2A). Three weeks later, mice were recorded continuously across each day of the estrous cycle. The GCaMP signals recorded from RP3V^{KISS} neurons (N=8) showed minimal fluctuations during metestrus and diestrus but became markedly more active on the afternoon of proestrus displaying an oscillatory pattern of increased baseline fluorescence associated with high frequency transients (Fig.2B,C). This heightened activity gradually subsided by the morning of estrus before returning to a quiescent state (Fig.B,C). An assessment of relative activity across the estrous cycle using area under the curve (AUC) demonstrated a significant increase in GCaMP fluorescence on proestrus compared to all of the other stages (Friedman Test: $\chi^2=16.2$, $p=0.0010$; Dunn's multiple comparisons tests: $p=0.0402$ (metestrus), $p=0.006$ (diestrus), $p=0.0402$ (estrus); Fig.2D). All five mice were found to have optic fibers running immediately alongside and lateral to GCaMP-expressing cells in the AVPV/PeN. Two mice with misplaced optic fibres exhibited almost no baseline activity (Fig.2E,F).

RP3V^{KISS} neuron population activity in relation to the LH surge on proestrus

To examine the relationship between the increased activity of RP3V^{KISS} neurons and the LH surge on proestrus, tail-tip bleeding was undertaken at 2–4-hour intervals for 8–10 hours (N=5). Peak LH levels (18.0 ± 3.2 ng/ml) occurred 3.5 ± 0.4 hours after the first RP3V^{KISS} neuron oscillation was detected with oscillations then continuing for a further 7.4 ± 1.6 hours after the decline in LH levels (Fig.3A,B).

Oscillating baseline shifts in RP3V^{KISS} neuron population activity on proestrus

To provide a detailed analysis of the GCaMP signals recorded on proestrus, a customised MATLAB code was used to separate slow and fast signal components. A 30-minute moving average was applied to the original dF/F signal (Fig.4A,B i) to extract the slow baseline changes in GCaMP activity (Fig.4A,B ii). Subtraction of this baseline from the original signal leaves the high frequency transients (Fig.4A,B iii).

The baseline shifts in GCaMP fluorescence exhibited an oscillatory pattern on proestrus in all 8 mice (Fig.4A,B ii & Fig.5C). The first identified oscillation was observed to begin 3.7 ± 0.5 hours (range: 1.6–5.3 hours) before lights-off with the total duration of oscillatory behaviour lasting 12.7 ± 0.7 hours (range: 10.0–15.0 hours) (Fig.5C,E). The mean duration of each slow oscillation was 90.8 ± 4.4 minutes (range: 39.3–166.1 minutes). Oscillations were occasionally observed in other stages of the cycle (3 of the 8 mice) and these had similar durations (metestrus: 88.6 ± 9.5 minutes, diestrus: 120.4 ± 29.7 minutes, estrus: 91.9 ± 21.7 minutes; Kruskal-Wallis test: $\chi^2=1.65$, $p=0.65$). The numbers of oscillations observed on proestrus were substantially greater than at all other stages of the cycle (Friedman Test: $\chi^2=18.75$, $p=0.0003$; Dunn's multiple comparisons tests: $p=0.0402$ (metestrus), $p=0.0029$ (diestrus), $p=0.0117$ (estrus); Fig.5E) and had significantly higher amplitudes (Friedman Test: $\chi^2=17.86$, $p=0.0005$; Dunn's multiple comparisons tests: proestrus 7.2 ± 0.9 ; metestrus 1.9 ± 1.0 $p=0.0221$, diestrus 1.1 ± 0.8 $p=0.0084$, estrus 1.5 ± 0.8 $p=0.0084$). Interestingly, a consistent ~90-minute oscillation started one hour preceding “lights-off” throughout the estrous cycle (Fig.5F).

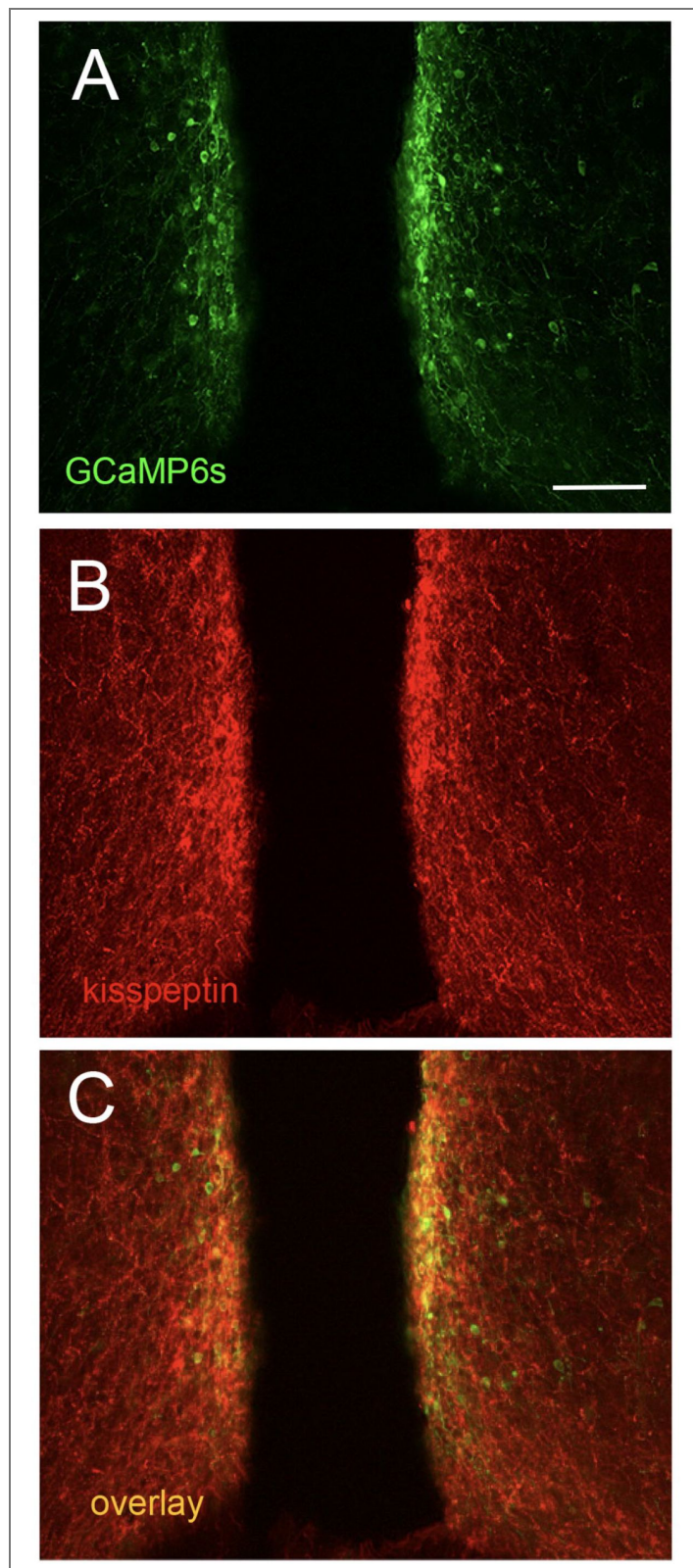


Figure 1. GCaMP expression in RP3V^{KISS} neurons.

Confocal images of **(A)** GFP (GCaMP6s, green) and **(B)** kisspeptin (red) and immunofluorescence in the RP3V, and **(C)** an overlay of both in the periventricular nucleus of a proestrous female mouse. Scale bar = 100 μ m.

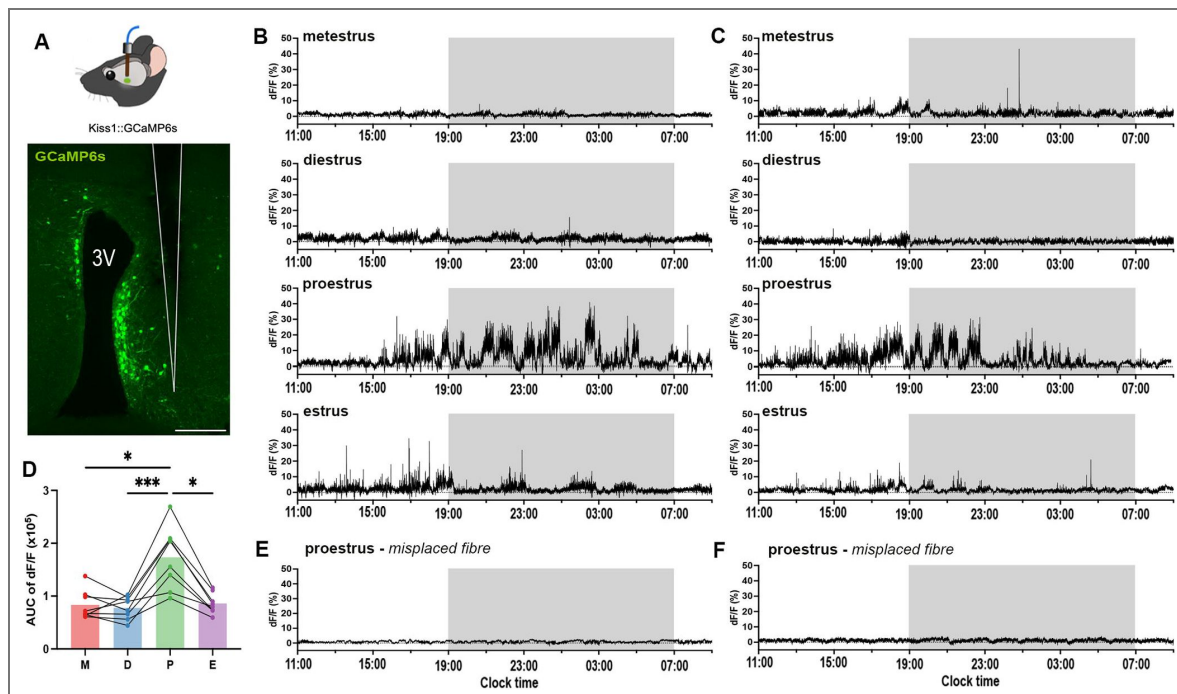


Figure 2. GCaMP signals in RP3V^{KISS} neurons across the estrous cycle.

(A) Coronal section showing location of tapered optic fibre (white outline) in relation to GCaMP-expressing cells (green) lining the third ventricle (3V). Scale bar = 100 μ m. (B&C) Two representative examples of 22-hour GCaMP fibre photometry recordings across the complete estrous cycle in two mice. Light-off period is represented by grey shaded area. (D) Mean area under the curve (AUC) calculated from 22-hour recordings across the estrous cycle: metestrus (M), diestrus (D), proestrus (P), and estrus (E). Each dot represents an animal (N=8). Friedman test followed by Dunn's multiple comparisons tests. * $p < 0.05$, *** $p < 0.001$. (E&F) Examples of 22-hour GCaMP fibre photometry recordings from two proestrous female mice with misplaced optic fibers in which signals remain < 3% of baseline.

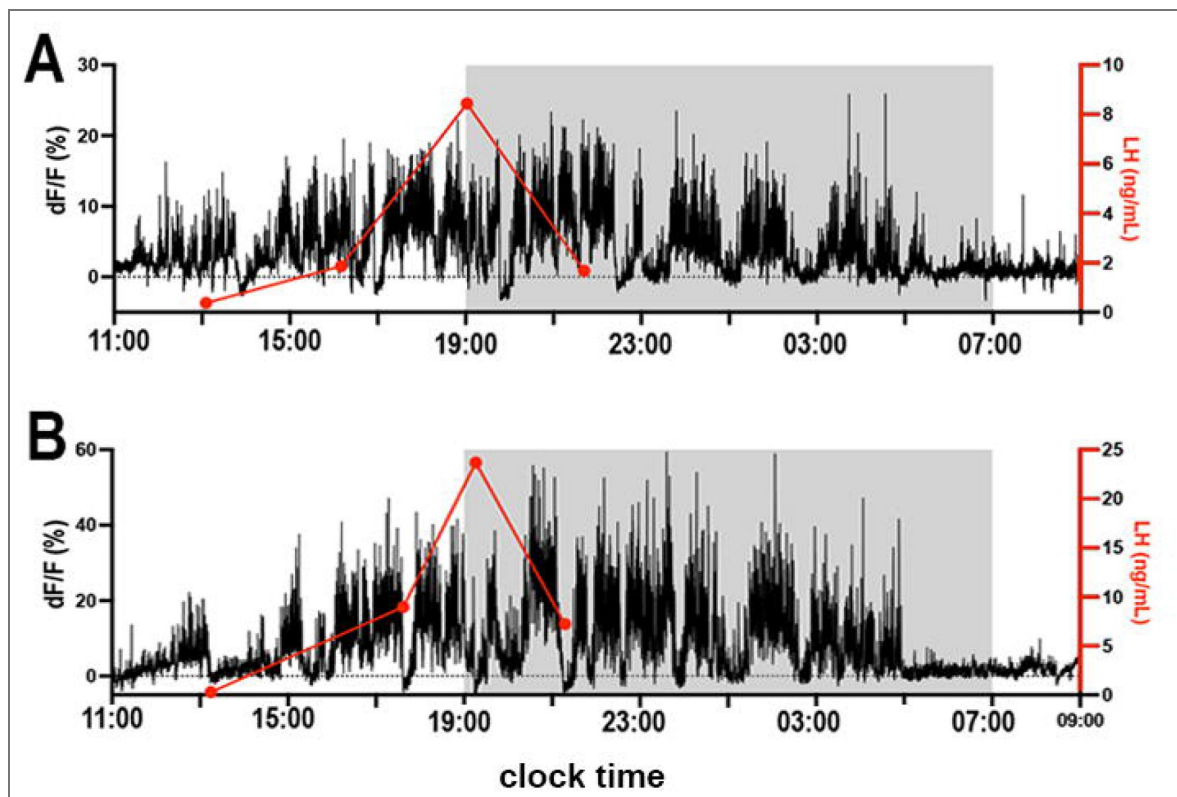


Figure 3. Relationship of RP3V^{KISS} neuron activity to the LH surge on proestrus.

(A,B) Two representative examples showing the relationship between the increase in GCaMP activity (black) with luteinizing hormone (LH) surge (red) assessed with 2-4h tail tip blood samples. Light-off period is represented by grey shaded area.

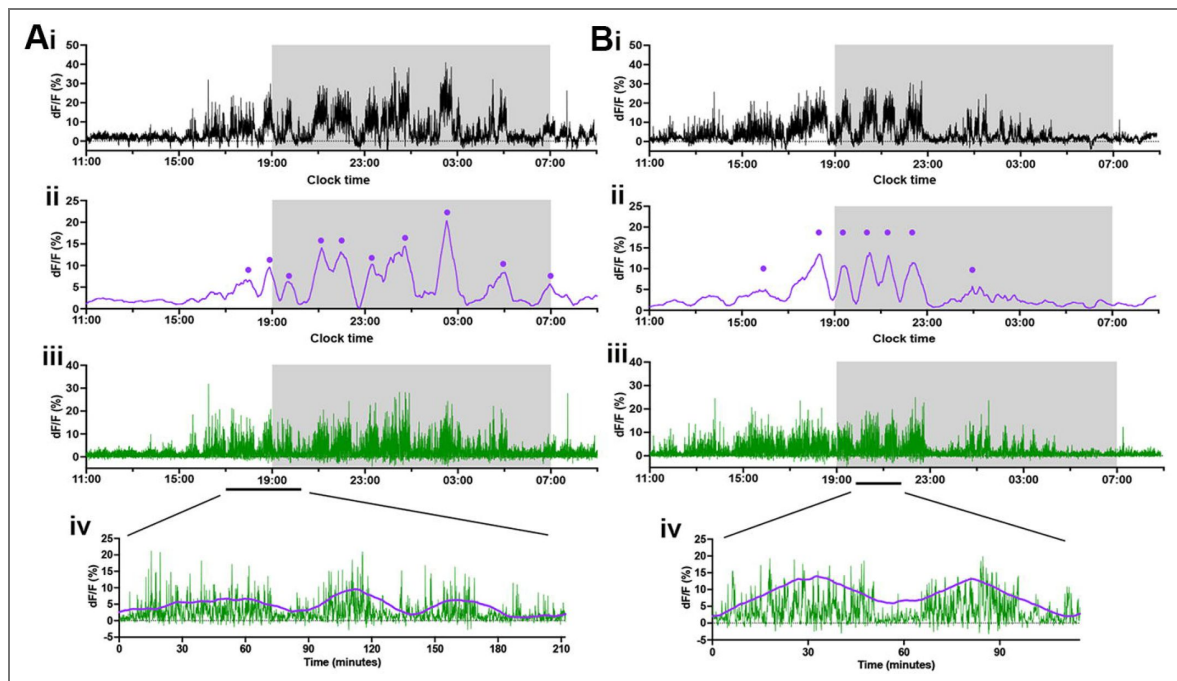


Figure 4. Deconvolution of RP3V^{KISS} neuron GCaMP signals

(A,B) Representative examples of 22-hour photometry recordings in proestrus from the same two female mice in Figure 2 showing (i) the original recording, (ii) a 30-minute moving average highlighting the baseline oscillations (purple) with identified oscillations labelled with dots, (iii) high frequency transients (green) after the moving average is subtracted from the original recording, and (iv) expanded views of the traces showing moving average and high frequency transients. Light-off period is represented by grey shaded area.

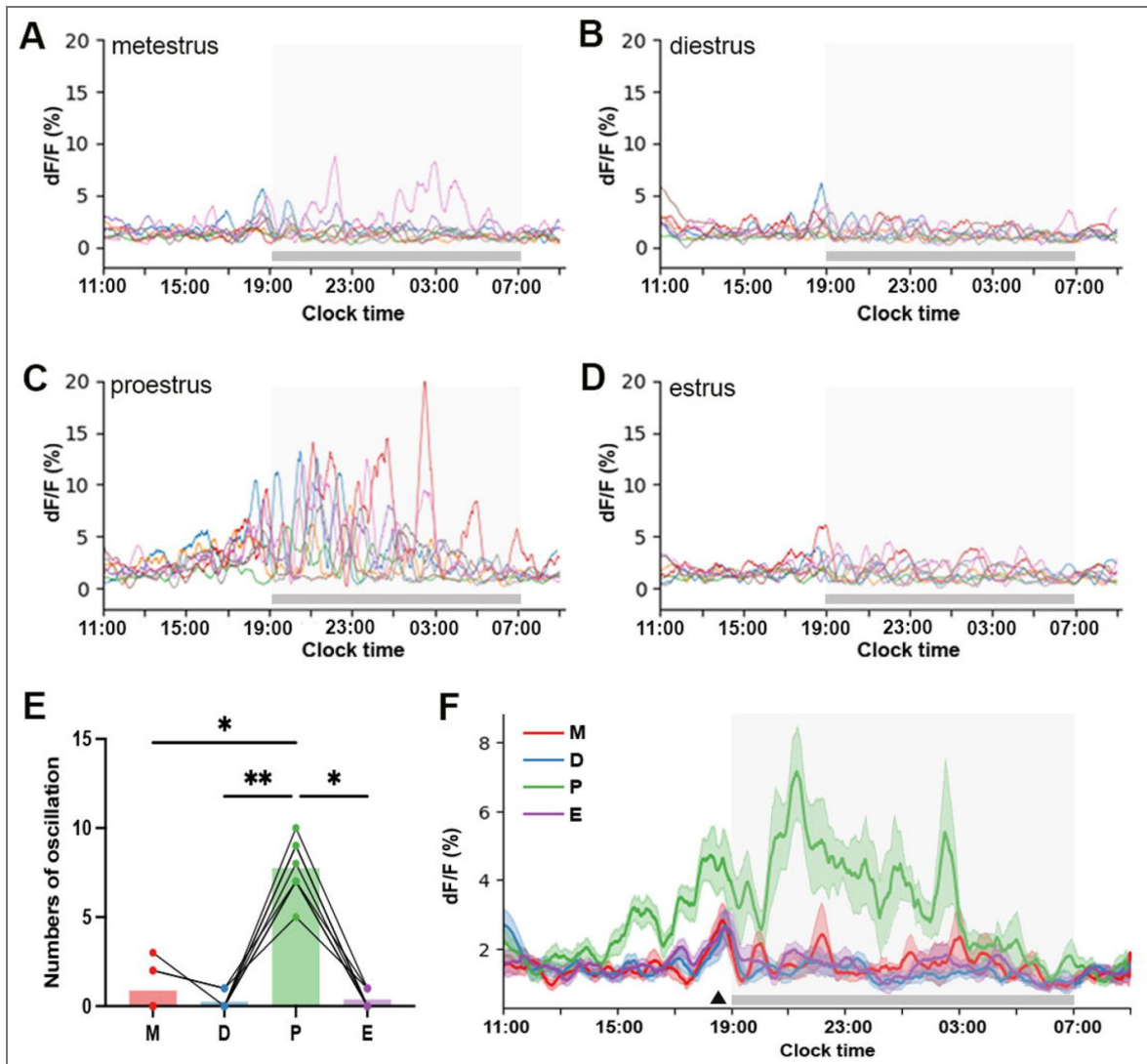


Figure 5. Baseline oscillations of RP3V^{KISS} neurons during proestrus.

(A–D) Individual baseline oscillation traces from 22-hour recordings for each animal across each of the estrous cycle stages (n=8). (E) Mean number of oscillations identified across each stage of the estrous cycle. Each dot represents an animal (Friedman Test: $\chi^2=18.75$, $p=0.0003$; Dunn's multiple comparisons tests versus proestrus: $p=0.0402$ (metestrus), $p=0.0029$ (diestrus), $p=0.0117$ (estrus). (F) Mean baseline oscillations showing variations across the estrous cycle: metestrus (M, red), diestrus (D, blue), proestrus (P, green), and estrus (E, purple). Shaded regions around the traces indicate $\pm$ SEMs of corresponding colours (N=8). Triangle marks the oscillation occurring before lights-off. Light-off period is represented by grey shaded area.

High frequency transient activity in RP3V^{KISS} neuron population activity

After baseline correction, high frequency transients were identified using a z-scoring approach, where dF/F traces were normalised and thresholded across multiple values ($k = 1$ to 4, step 0.2). This method provides a data-driven, threshold-independent way to detect transients while reducing the impact of noise and ensuring robustness across different signal amplitudes.

Transient activity was detected throughout the estrous cycle but had its highest frequency on proestrus (Repeated measures one-way ANOVA: $F=24.35$, $p<0.0001$; Tukey's multiple comparisons tests: $p=0.0025$ (metestrus), $p=0.0003$ (diestrus), $p=0.0008$ (estrus); Fig.6A,B). Similarly, the amplitude of individual transients was higher in proestrus compared to other stages (Repeated measures one-way ANOVA: $F=35.58$, $p=0.0001$; Tukey's multiple comparisons tests: $p=0.0032$ (metestrus), $p=0.0010$ (diestrus), $p=0.0014$ (estrus); Fig.6A,C). Each transient typically had a duration of ~ 10 s and this changed slightly across the cycle being longer in proestrus compared to estrus (Repeated measures one-way ANOVA: $F=7.71$, $p=0.0059$; Tukey's multiple comparisons tests: $p=0.0053$), with transients also being longer in metestrus when compared to diestrus (Tukey's multiple comparisons tests: $p=0.040$) (Fig.6D). When aligning high frequency transients with baseline oscillations, the up-state of each oscillation appears to coincide with an increased frequency of transients, whereas the troughs in the baseline correlate with a slower frequency of transients (Fig.4A,B iv).

Shifting onset of RP3V^{KISS} neuron population activation across subsequent proestrous stages

The onset of GnRH neuron surge activity exhibits substantial variability between mice and also within cycles in individual mice (Han, Yeo et al. 2025). To examine the consistency of the surge-like behaviour observed in proestrus, RP3V^{KISS} neuron activity was recorded across two or three proestrous stages in the same mouse and GCaMP signals were deconvolved to assess baseline shifts.

All seven mice exhibited variability in the initiation and duration of oscillatory activity occurring during proestrus across different cycles (Fig.7A,B). On average, the peak of the first oscillation varied by 2.0 ± 0.5 hours when examining surge activity in the same mice (range: 0.4-4.2 hours; $n=7$). There was no consistent pattern in surge onset timing across subsequent proestrous surges with the onset fluctuating both forward and backward (Fig.7A,B).

RP3V^{KISS} neuron population activity in OVX, estrogen-replaced female mice

Rising follicular phase levels of estradiol (E2) are the key signal driving the GnRH surge generator. To examine the effect of estradiol on RP3V^{KISS} neuron activity, five of the mice reported above were ovariectomised (OVX) and subjected to a well-established estrogen replacement regimen involving the placement of E2-containing Silastic capsules followed by s.c. injection of estradiol benzoate (EB) (Bronson 1981) (Fig.8). This also provided within-animal comparisons between intact and OVX states as well as the proestrous and OVX+E2+EB-induced surges.

Mice examined one week after OVX exhibited a significant decrease in RP3V^{KISS} neuron population activity compared with the intact (diestrus) state (Figs.8A,B & 9A; Paired t-test: $p=0.002$). Despite an upward trend, the subsequent replacement of E2 did not result in any significant change in RP3V^{KISS} neuron activity 5 days after capsule implantation (Figs.8B,C & 9B) (Friedman test: $\chi^2=10.68$, $p=0.0055$ for all groups; Dunn's multiple comparisons tests: $p=0.26$ (OVX vs OVX+E2)). This procedure has been shown previously to return estradiol to physiological diestrus levels (Porteous, Haden et al. 2021). Similarly, the injection of EB on day 6 did not have any significant effect on GCaMP signal in OVX+E2+EB mice (Fig.8C,D & 9B) Dunn's multiple comparisons tests: $p>0.99$ (OVX vs OVX+E2+EB). However, the following day, when an LH surge

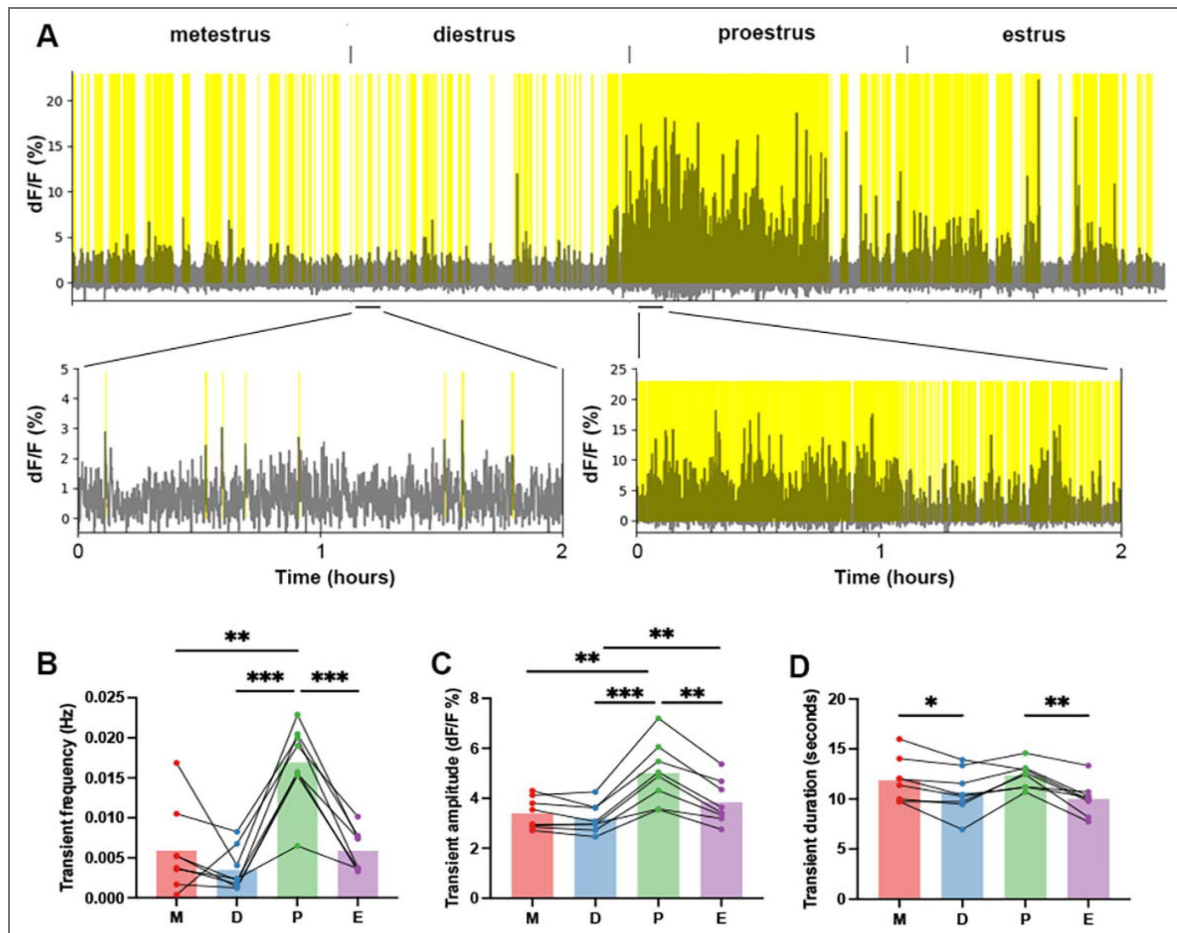


Figure 6. Changes in RP3V^{KISS} neuron high frequency transients across the cycle.

(A) An example of high frequency transient signals across the estrous cycle in one mouse. Identified significant transients are highlighted in yellow and expanded views given below. Note different y-axes for diestrus and proestrous graphs. (B) Mean frequency (Hz), (C) amplitude (dF/F, %) and (D) duration (seconds) of transients in metestrus (M), diestrus (D), proestrus (P), and estrus (E). Each dot represents an animal (N=8). Repeated measures one-way ANOVA followed by Tukey's multiple comparisons test; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (see text for exact p values).

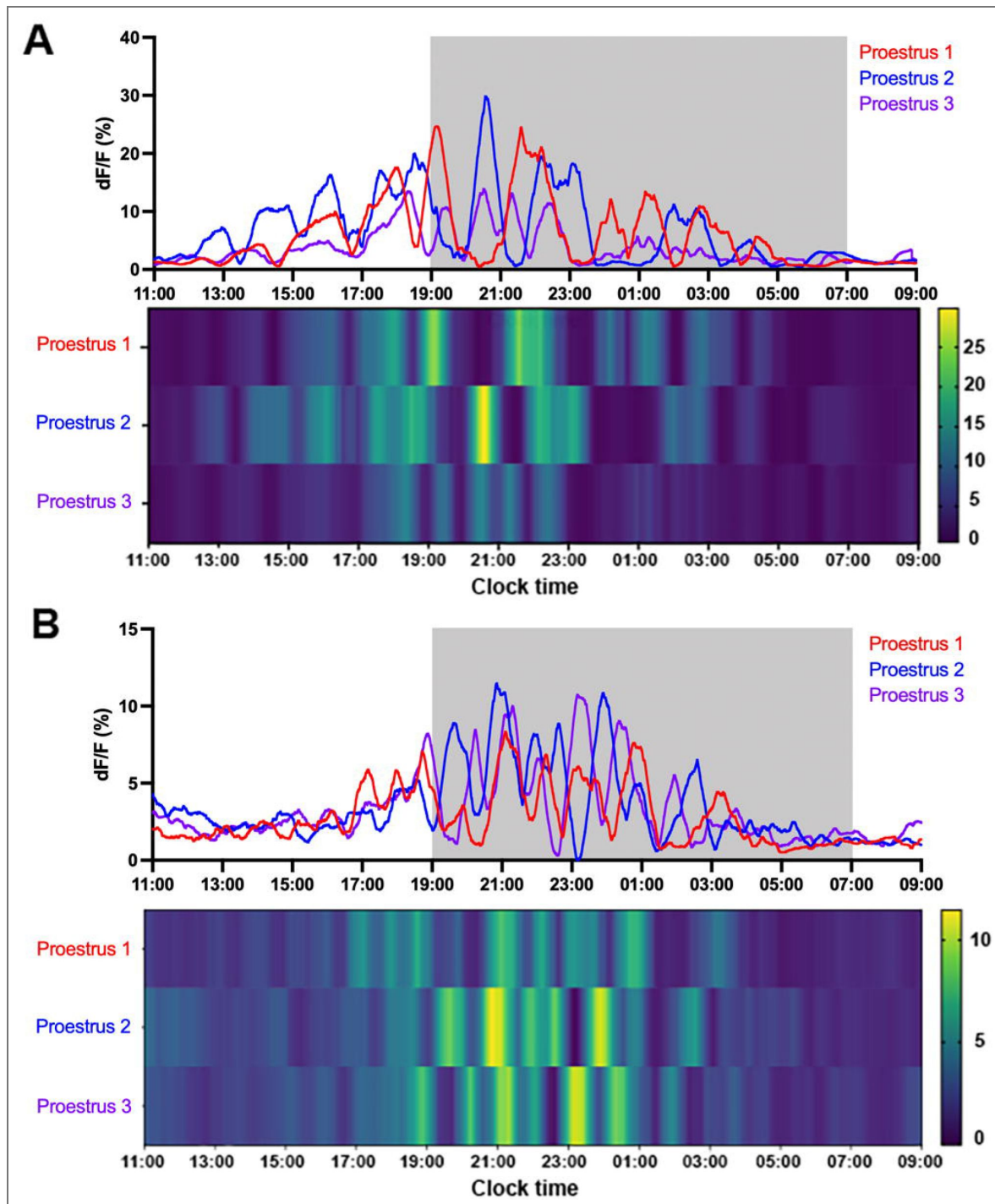


Figure 7. Variable oscillations in RP3V^{KISS} neuron activity across proestrous surges in the same mice.

(A&B) Two representative examples showing code-extracted baseline profiles of RP3V^{KISS} neuron GCaMP activity across three proestrous surges in the same mouse. Light-off period is represented by grey shaded area. Heat maps below show calcium signal dynamics (dF/F) with each row representing one proestrus. Colour intensity indicates dF/F values, with yellow and green colours representing higher signal amplitudes. Legends of heat maps show colour codes for dF/F (%).

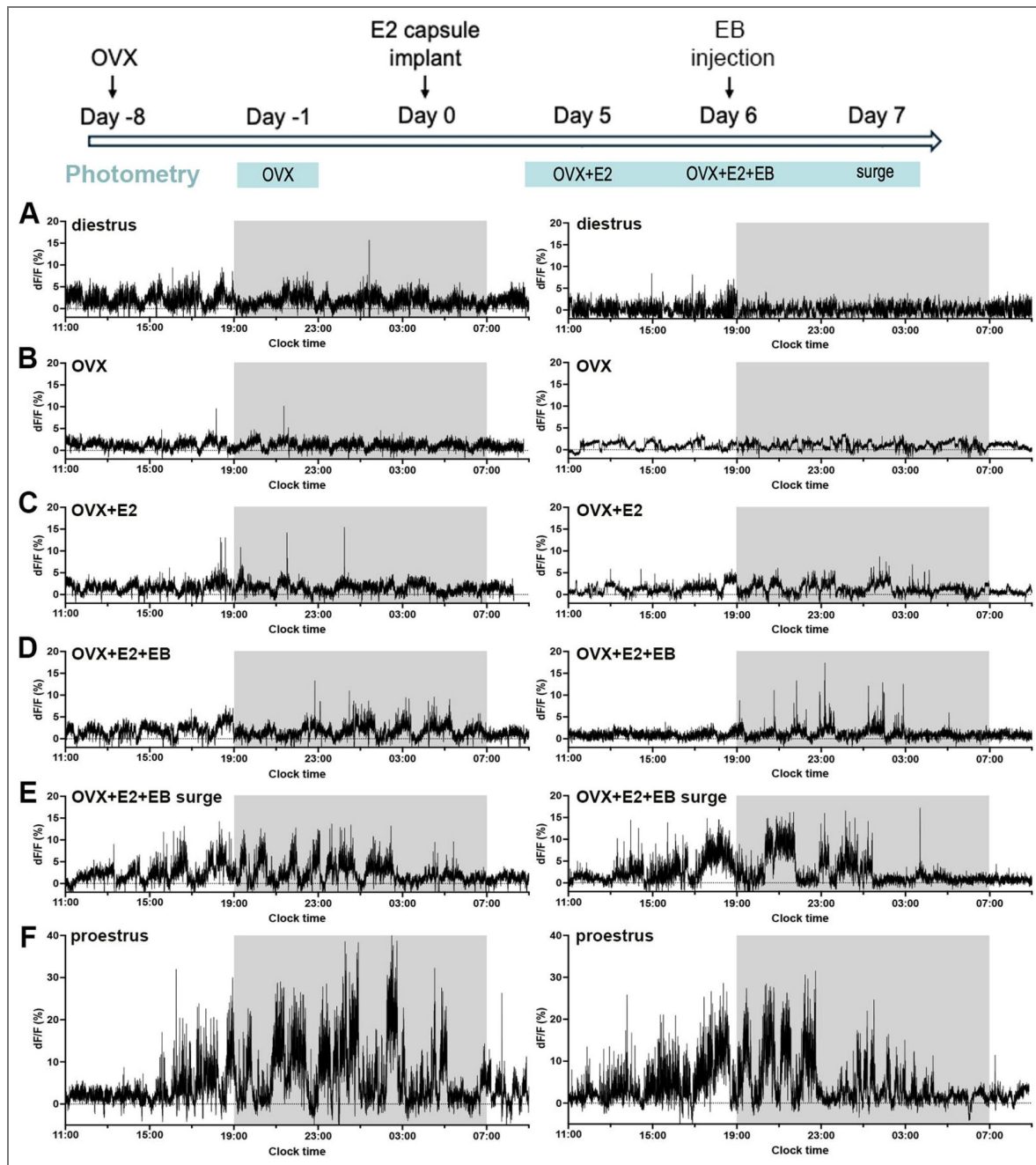


Figure 8. Effects of ovariectomy and estrogen replacement on RP3V^{KISS} neuron activity.

Experimental plan is shown at the top with times of photometry recordings highlighted in blue. Representative examples of 22-hour photometry recordings from two female mice (right and left) in (A) diestrus, (B) one week after OVX, (C) five days after the estradiol implant (OVX+E2), (D) on the day of estradiol benzoate injection (OVX+E2+EB), (E) at the time of the expected estradiol-induced LH surge (OVX+E2+EB surge), and for comparison (F) in proestrus (before ovariectomy). Light-off period is represented by grey shaded area.

was expected, RP3V^{KISS} neuron activity became elevated with the appearance of baseline oscillations and enhanced transient activity (Fig.8E) with a significant increase in AUC fluorescence compared with OVX mice (Dunn's multiple comparisons tests: $p=0.021$; Fig.8B). Qualitatively, the pattern of RP3V^{KISS} neuron surge activity appeared similar on proestrus and OVX+E2+EB in the same mice (Fig.8E,F) but the magnitude of change was substantially reduced in the OVX+E2+EB condition (paired t-test: $p=0.014$; Fig.9C).

To evaluate RP3V^{KISS} neuron activity in more detail in the OVX+E2+EB paradigm, the same deconvolution methods employed for the intact recordings were used (Fig.9D,E). Three of five females exhibited identified oscillations in the OVX+E2+EB surge state with 3.7 ± 0.3 oscillations (range: 3-4 oscillations) occurring during the recording period and these had a duration of 88.6 ± 10.1 minutes per oscillation (range: 59.7-169.3 minutes), and an amplitude of $4.5 \pm 1.2\%$ (range: 2.2-6.1%). Average total duration of oscillatory activity in these mice was 7.1 ± 0.5 hours (range: 6.1-8.0 hours); significantly shorter compared to that of proestrus (12.7 ± 0.7 hours; unpaired t-test, $p=0.001$). During the other treatment conditions, one out of five females exhibited a single identified oscillation on the day of EB injection (OVX+E2+EB).

Notably, the oscillation observed ~1h before lights-off in intact mice disappeared after OVX but appeared to be restored under all estrogen replacement conditions (Fig.9D).

The high frequency transients (Fig.9E) were evaluated across all four conditions. The frequency of transients was significantly elevated during the OVX+E2+EB surge condition compared to OVX (Friedman test: $\chi^2=14.04$, $p<0.0001$; Dunn's multiple comparisons tests: $p=0.0014$) (Fig.8E, Table 1). The durations of transients were not different across conditions (Table 1, Repeated measures one-way ANOVA: $F=0.50$, $p=0.53$). Although repeated measures one-way ANOVA revealed a significant main effect of estrogen treatment on transient amplitude ($F=8.1$, $p=0.039$), post-hoc analysis using Tukey's multiple comparisons tests did not detect any significant pairwise differences between individual conditions (Table 1).

Discussion

We report here that RP3V^{KISS} neurons exhibit a marked increase in population activity on the afternoon of proestrus that consists of ~90-min duration oscillations, associated with high frequency transient activity, that last for approximately 13 hours. The first half of this activation period almost certainly drives the proestrous LH surge (Goodman, Herbison et al. 2022, Kauffman 2022) while the role of the latter activity remains less certain. We show that this behavior of RP3V^{KISS} neurons is driven by circulating estradiol although the standard OVX+E2+EB model of estrogen replacement used here appears insufficient to drive the normal magnitude of RP3V^{KISS} neuron activity.

The activity of the RP3V^{KISS} neuron population is relatively stable during metestrus and diestrus with only very occasional excursions from basal activity. This changes markedly on the afternoon of proestrus when substantial baseline oscillations begin and are recorded for the next ~ 13 hours. While this change in RP3V kisspeptin neuron activity almost certainly drives the LH surge, the precise temporal relationship between the initial onset of RP3V^{KISS} neuron activity and LH secretion remains unclear. The resolution of blood sampling (hours) was poor compared to that of the neural recordings (seconds), and insufficient to establish any precise correlation. The abrupt activation of GnRH neurons by exogenous kisspeptin (Han, Gottsch et al. 2005) or LH secretion by optogenetic activation of RP3V^{KISS} neurons (Piet, Kalil et al. 2018) indicates that LH secretion would likely be elevated soon after any increment in kisspeptin transmission. Indeed, the in vivo temporal profiles of RP3V kisspeptin neurons and GnRH neuron dendrons are near identical across proestrus suggesting a high degree of temporal fidelity (Fig.10). However, more detailed LH sampling protocols will need to be implemented alongside GCaMP fibre photometry to examine the precise relationship between the start of RP3V kisspeptin neuron activity and LH secretion.

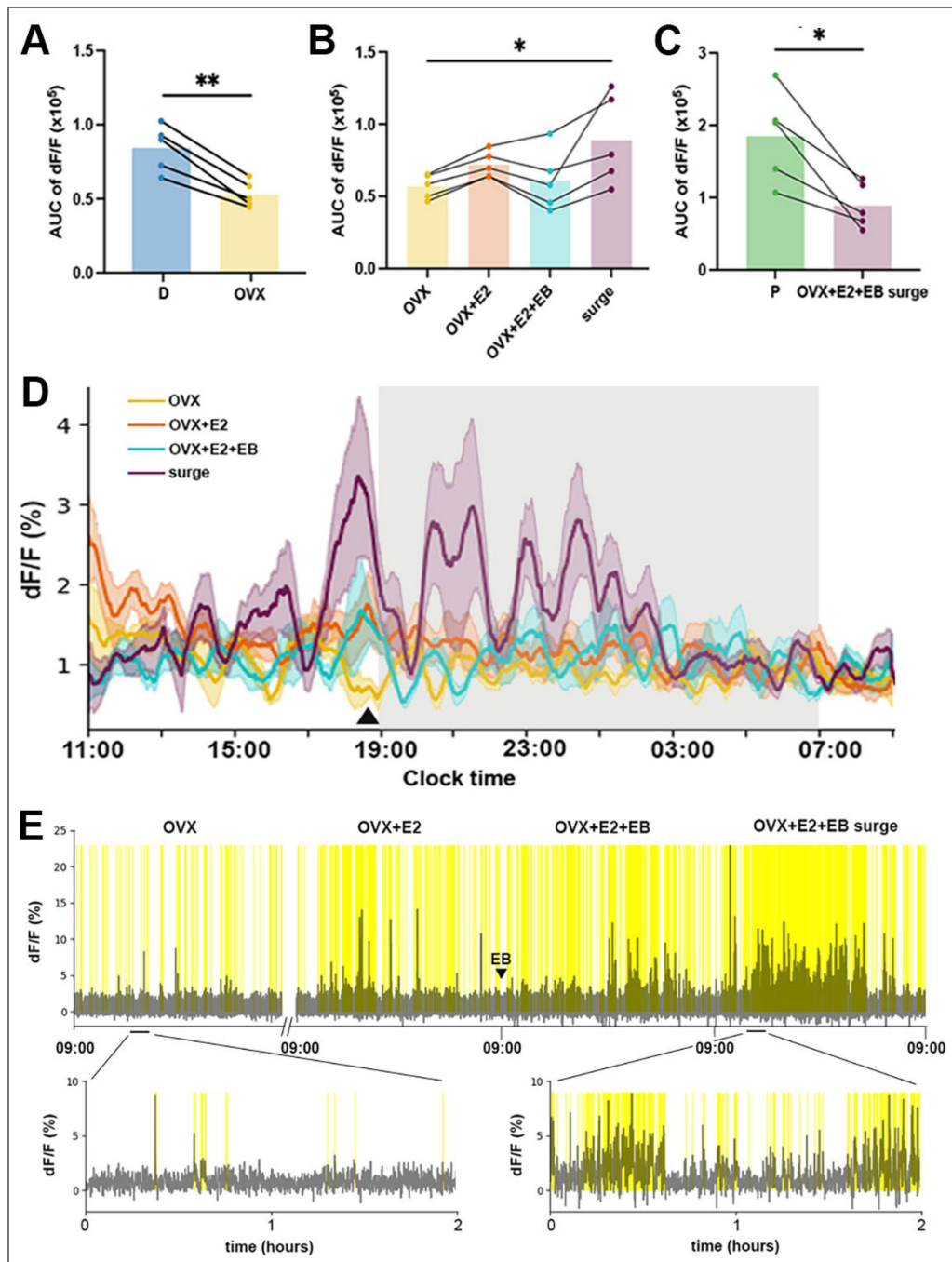


Figure 9. Changing RP3V^{KISS} neuron baseline and transient activity following OVX + estrogen treatment regimens.

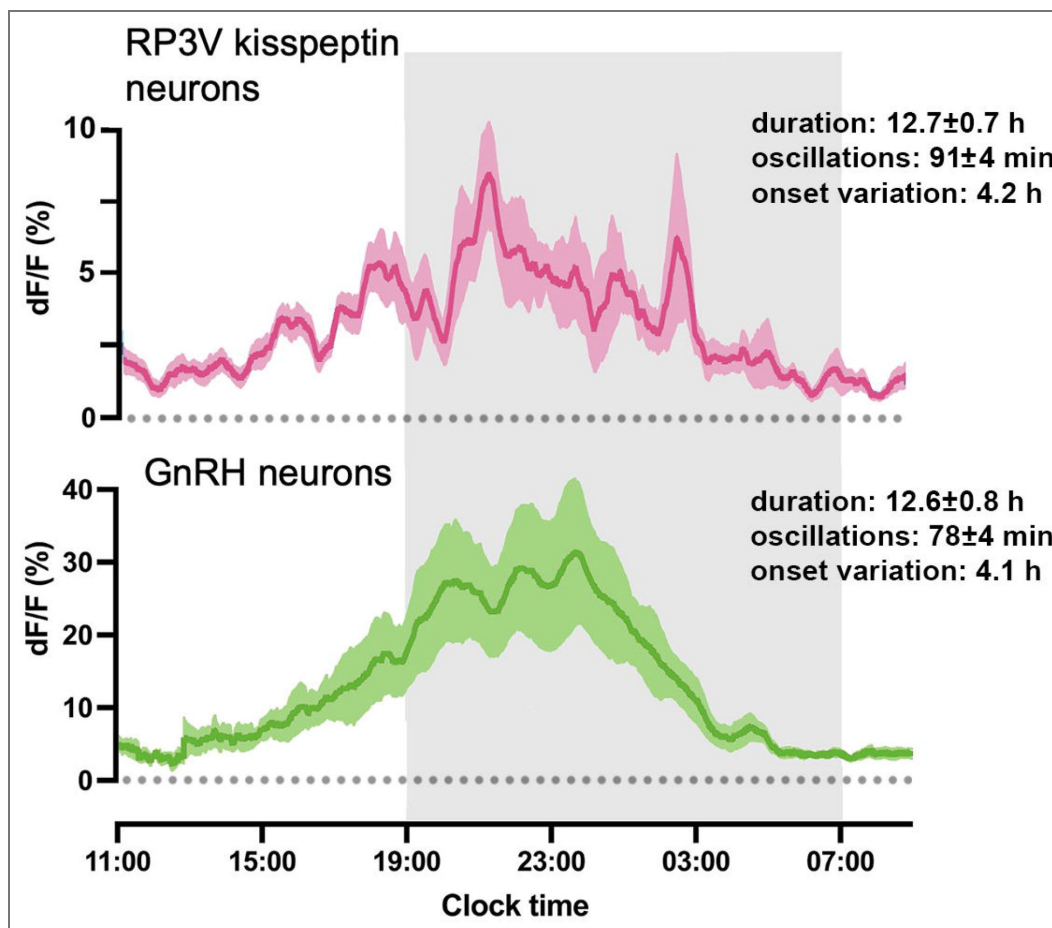
(A) Mean area under the curve (AUC) of 22-hour recordings for mice in diestrus (D) and following their ovariectomy (OVX), paired t-test: $p=0.002$ (B) Mean AUC calculated in OVX, OVX+E2, OVX+E2+EB, and OVX+E2+EB surge conditions. Friedman test: $\chi^2=10.68$, $p=0.0055$, Dunn's multiple comparisons test $p=0.021$. (C) Mean AUC from intact proestrous mice (P) compared to the same mice under OVX+E2+EB surge conditions, paired t-test: $p=0.014$. Each dot represents one animal (N=5). (D) Mean moving average of baseline recordings over 22-hour recordings showing variations after OVX (yellow), five days after estradiol implant (OVX+E2, orange), on the day of EB injection (OVX+E2+EB, aqua), and on the day of OVX+E2+EB surge (plum). Shaded regions around the traces indicate SEMs of corresponding colours (n=5). Triangle marks the pre-lights-off oscillation. Light-off period is represented by grey shaded area. (E) Example of high frequency transient activity from a representative mouse 7 days after OVX and then continuously for three days encompassing the fifth day after E2 implant (OVX+E2), on the day of EB injection (OVX+E2+EB), and the day of the expected OVX+E2+EB LH surge. Triangle indicates time of EB injection. Identified significant transients are highlighted in yellow. Expanded views of the traces are shown below.

Table 1. High frequency transients parameters in different treatment groups (Mean $\pm$ SEM)

Treatment	Frequency (Hz)	Duration (seconds)	Amplitude (dF/F %)
OVX	0.0020 ± 0.00046	14.36 ± 1.85	2.30 ± 0.22
OVX+E2	0.0038 ± 0.00028	14.34 ± 0.86	2.47 ± 0.26
OVX+E2+EB	0.0044 ± 0.00029	13.40 ± 0.79	2.55 ± 0.27
OVX+E2+EB surge	0.0120 ± 0.00190	13.13 ± 1.00	3.03 ± 0.39

Figure 10. Comparison between RP3V^{KISS} and GnRH neuron activity patterns on proestrus.

Mean 30-minute moving average of 22-hour GCaMP recordings across proestrus from RP3V^{KISS} neurons (pink; taken from Fig. 4F; N=8) and GnRH neuron dendron activity (green; raw data obtained from (Han, Yeo et al. 2025); n=7) processed in the same manner. Shaded regions around the traces indicate $\pm$ SEMs of corresponding colours. Text provides mean $\pm$ SEM duration of total elevated activity, duration of each oscillation, and variation in onset of oscillations between animals.



We find that RP3V^{KISS} neuron activity continues for many hours past the peak of the LH surge. This is the same as the profile of GnRH neuron activity in the mouse on proestrus (Han, Yeo et al. 2025 [DOI](#), Liu, Shen et al. 2025 [DOI](#)) (Fig.10 [DOI](#)) and long-standing observations that GnRH secretion at the median eminence long outlast the LH surge in multiple species (Sarkar, Chiappa et al. 1976 [DOI](#), Pau, Berria et al. 1993 [DOI](#), Karsch, Bowen et al. 1997 [DOI](#)). While it remains unclear what the role of GnRH secretion after the LH surge may be (Skinner and Caraty 2002 [DOI](#)), it is very likely that the extended period of GnRH and LH secretion is driven by prolonged RP3V^{KISS} neuron input to the GnRH neurons.

The impact of prolonged RP3V^{KISS} neuron activity on other neural networks in the forebrain during proestrus remain unknown. The RP3V^{KISS} neurons project widely throughout the limbic forebrain (Yeo and Herbison 2011 [DOI](#)) and may conceivably operate to coordinate multiple functions relevant to reproduction at proestrus. Indeed, commensurate with the long period of elevated RP3V^{KISS} neuron activity detected here following the surge, these neurons have been proposed to modulate female mate preference and copulatory behavior (Hellier, Brock et al. 2018 [DOI](#)) that would typically occur a few hours after the surge. At present, it remains unknown whether the sub-population of RP3V^{KISS} neurons innervating GnRH neurons also send collaterals to other brain regions.

The recent ability to record the activity of the GnRH neuron population in vivo revealed that their elevated activity on the afternoon of proestrus occurred in an oscillatory manner with a period of ~ 78 min (Han, Yeo et al. 2025 [DOI](#)). This provides a striking parallel with the ~ 91 min-duration oscillations now detected in RP3V^{KISS} neurons activity over the exact same time period. This suggests that the unusual oscillatory pattern of GnRH neuron activity on proestrus is being driven by RP3V^{KISS} neurons. Indeed, the entire profile of RP3V^{KISS} and GnRH neuron activity across proestrus is strikingly similar (Fig.10 [DOI](#)) making it very likely that GnRH neuron activity on proestrus is patterned almost entirely by the RP3V^{KISS} neurons. Nevertheless, a small role may also exist for the ARN kisspeptin neuron pulse generator as it remains active during the ascending phase of the proestrous LH surge (McQuillan, Han et al. 2019 [DOI](#), Han, Yeo et al. 2025 [DOI](#)). As such, the episodic release of kisspeptin from the ARN pulse generator on GnRH neuron dendrons may potentiate early GnRH secretion driven by RP3V^{KISS} neurons. However, the functional impact of ARN kisspeptin neurons on the surge remains unclear in mice. The deletion of ARN kisspeptin neurons was not found to have any effect on the LH surge (Coutinho, Esparza et al. 2024 [DOI](#)) while selective removal of kisspeptin from ARN neurons resulted in a decrease in the amplitude of the LH surge (Velasco, Franssen et al. 2023 [DOI](#)).

It is interesting to speculate on how oscillatory activity may be generated in RP3V^{KISS} neurons and what role it may play. We find that the baseline GCaMP oscillations are associated with intense high frequency transient activity with each transient having a duration of approximately 10 s. This may represent the signature of single or small groups of RP3V^{KISS} neurons exhibiting 10 s burst firing that then go on to synchronise into 90 min episodes of activity. In acute brain slices, RP3V^{KISS} neurons often exhibit spontaneous burst firing (Jamieson and Piet 2022 [DOI](#)) and each burst is known to generate 5-10 s-duration calcium transients in GCaMP recordings (Piet, Fraissenon et al. 2015 [DOI](#)). It is also interesting to note that the optogenetic induction of burst firing in RP3V^{KISS} neurons is the most efficient mode for activating GnRH neurons and LH secretion (Piet, Kalil et al. 2018 [DOI](#)). How RP3V^{KISS} neurons may synchronise their activity is unknown. It is possible that, like ARN kisspeptin neurons, they use direct recurrent collateral innervation to facilitate periods of synchronous bursting. Alternatively, synchronicity may be generated within a wider network of RP3V circuitry and/or by afferent inputs. The physiological roles and necessity for episodic activity in GnRH neurons, driven by RP3V^{KISS} neurons, are also unknown. As these neurons are intensely active for prolonged periods of over 12 hours, it is possible that recurrent episodes of inhibition protect RP3V^{KISS} and GnRH neurons from excitotoxicity.

It is well established that the onset of the LH surge is entrained by circadian inputs in rodents (Tonsfeldt, Mellon et al. 2022 [DOI](#), Piet 2023 [DOI](#)). However, this entrainment is not highly time locked compared to other circadian-timed events such as locomotion (Starnes and Jones 2023 [DOI](#)). We show here that the onset of RP3V^{KISS} neuron activity between mice, and even within the same

mouse, varies by several hours around the time of “lights off” on proestrus. The exact same observations have been made for GnRH neuron activation and the LH surge on proestrus (Miller, Olson et al. 2004 [↗](#), Minabe, Uenoyama et al. 2011 [↗](#), Czielesky, Prescott et al. 2016 [↗](#), Yeo, Han et al. 2025 [↗](#)). The mechanisms responsible for this variability are not known. Although only three of the OVX+E+E mice exhibited RP3V^{KISS} neuron oscillations, they also initiated at different times suggesting perhaps that the variability is not due to differences in the time estrogen peaks.

Given this variability, we were surprised to observe a tightly time-locked RP3V^{KISS} neuron increment in activity almost exactly one hour before “lights off” on proestrus, as well as on every other day of the cycle. While this would appear to be a more rigid circadian input to the RP3V^{KISS} neurons, its role in triggering the LH surge is unclear and will require further investigation. Interestingly, OVX mice were the only group in which we did not observe this circadian event. This dependence on estrogen is reminiscent of the excitatory effect of vasopressin on RP3V^{KISS} neuron firing that is unchanging in different estrogenic states but absent in OVX mice (Piet, Fraissenon et al. 2015 [↗](#)).

We find that the activity of the RP3V^{KISS} neuron population is reduced 7-days following ovariectomy but that relatively little change is evoked by a sequential regimen of estrogen replacement until the day of the expected surge when proestrus-like oscillatory activity emerges. This is consistent with the concept of the surge being generated by rising estrogen levels that harness slow transcriptional mechanisms to alter RP3V^{KISS} neuron excitability (Glidewell-Kenney, Hurley et al. 2007 [↗](#), Herbison 2015 [↗](#)). Increments in circulating estradiol either in the OVX+E model or during diestrus, when estradiol levels peak (Wall, Desai et al. 2023 [↗](#)), appear to have little immediate effect on RP3V^{KISS} neuron activity patterns. This is broadly in agreement with acute brain slice studies that have consistently found reduced RP3V^{KISS} neuron firing rates following OVX but then rather few consistent changes in firing across the estrous cycle itself (Jamieson and Piet 2022 [↗](#)). Nevertheless, it is clear that estrogen treatment of OVX mice slowly up-regulates sodium (I_{NaP}), calcium (I_T), and hyperpolarization-activated (I_h) ion channels in RP3V^{KISS} neurons, that would all be expected to result in increased excitability (Jamieson and Piet 2022 [↗](#)). Once established, these changes in ion channel expression are presumably at least partly responsible for the enhanced oscillatory-like activity patterns of RP3V^{KISS} neurons held in check until the day of the surge.

While the same pattern of RP3V^{KISS} neuron activity is observed on the afternoon of proestrus and in the OVX+E2+EB model, it is evident that the magnitude of change is dramatically reduced in the latter. We did not assess surge LH levels in the OVX+E2+EB mice so it is possible that the two animals that did not exhibit oscillatory activity did not surge. For those that did, it is likely that they exhibited LH surges with a lower amplitude compared to proestrus, as prior work from the laboratory using the same model has shown that peak LH surge levels in OVX+E2+EB mice (7.7 ± 0.7 ng/mL) are half of that observed in proestrus mice (14.5 ± 1.5 ng/mL) (Czielesky, Prescott et al. 2016 [↗](#)). This inability to recreate full proestrous-like LH surge levels is typical of estrogen-replaced OVX mouse models (Bronson and Vom Saal 1979 [↗](#), Dror, Franks et al. 2013 [↗](#), Silveira, Burger et al. 2017 [↗](#)). We would suggest that the reduced amplitude of the LH surge in estrogen-replaced OVX models likely originates from sub-optimal activation of the RP3V^{KISS} neurons. This may be due to the lack of sufficient progesterone or other ovarian factors in estrogen-replaced OVX models (Micevych and Sinchak 2011 [↗](#)). However, it is important to note that the seminal experiments of Bronson and Vom Saal (Bronson and Vom Saal 1979 [↗](#)) identified that a full LH surge can be achieved in OVX+E2+EB mice with careful attention to the timing of the EB injection. Our present OVX+E2+EB protocol matches the most efficacious protocol of Bronson and Vom Saal with the exception of using 4 compared to 5 μ g E2 capsules and a 12:12 rather than 14:10 lighting regimen.

In summary, we demonstrate that RP3V^{KISS} neurons exhibit an unusual, extended period of baseline oscillatory activity on proestrus. The genesis of oscillatory RP3V^{KISS} neuronal activity is critically dependent on long-term exposure to estradiol with a likely involvement of circadian inputs. This oscillatory RP3V^{KISS} neuron activity almost certainly drives the same pattern of GnRH neuron activity detected at this time which is critical for triggering the preovulatory LH surge. It

possible that oscillatory RP3V^{KISS} neuron activity existing beyond the generation of the LH surge operates to coordinate other neural circuits controlling reproduction such as those underlying female sexual behavior (Hellier, Brock et al. 2018 [DOI](#)).

Materials and Methods

Animals

Adult *B6(129S4)-Kiss1^{tm1.1(cre/EGFP)Rpa/J}* (Palmiter *Kiss1^{Cre/+}* (v2), JAX stock #033169) (Padilla, Johnson et al. 2018 [DOI](#)) female mice were group-housed in conventional cages with environmental enrichment under conditions of controlled temperature (22±2°C) and lighting (12-hour light/12-hour dark cycle; lights on at 07:00) with *ad libitum* access to water and low-phytoestrogen food (Tekland 2016, Envigo RMS, UK) and water. Following surgery, mice were single housed in conventional cages under the same conditions until the end of the study. All animal experimental protocols were approved by the University of Cambridge Animal Welfare and Ethics Review Body under UK Home Office licenses P174441DE and PP9818192.

Stereotaxic implantation of optic fibres and AAV injections

Adult female mice (10-14-weeks-old) were anaesthetised with 2% isoflurane and placed in a stereotaxic frame with buprenorphine (0.05mg/kg, s.c.) and meloxicam (5 mg/kg, s.c.) analgesia. Dexamethasone (10 mg/kg, s.c.) was used to prevent cranial swelling. A custom-built Hamilton syringe apparatus, holding a 5-μL Hamilton syringe (Hamilton, catalogue No. 7634-01) with a 29-gauge bevelled needle (Hamilton, catalogue No. 90029), was filled with 2.0 μL of recombinant Cre-dependent AAVs encoding GCaMP6s (AAV9CAG.FLEX.GCaMP6s.WPRE.SV40, Addgene 100842-AAV9, titre: 2 x 10¹³ genome copy/mL). Mice for photometry experiments received a single unilateral injection of 2.0 μL AAV at coordinates 0.8 mm anterior to bregma, 0.25 mm lateral to the superior sagittal sinus, and 5.0 mm deep. Mice for immunohistochemistry characterisation received bilateral AAV injections. After a unilateral viral injection, an indwelling tapered optical fibre (400 μm diameter, 0.66 NA, 3.5 mm taper length, Doric Lenses, Quebec, Canada) was implanted directly into the RP3V on the same side (0.55 mm anterior to bregma, 0.22 mm lateral to the superior sagittal sinus, 5.2 mm deep). One week following surgery, all animals were handled daily and habituated to a photometry recording set-up for at least three weeks. Estrous cycle stage was assessed by vaginal lavage each morning.

Estrogen-induced surge model

All estrogen-induced surge studies in the laboratory use the OVX+E2+EB protocol of Bronson and Vom Saal with slight modifications (Bronson and Vom Saal 1979 [DOI](#)). One week after OVX, mice were implanted subcutaneously with silastic implants (id, 1.02 mm; od, 2.16 mm, Dow Corning Corp) containing 4 μg 17β-estradiol (E2, Sigma-Aldrich, catalogue No. E8875) per 20 g body weight. This dose of 4 μg has been shown to return E2 levels to diestrous concentrations (Porteous, Haden et al. 2021 [DOI](#)). Six days after capsule implantation, mice were given a subcutaneous injection of 1 μg β-estradiol 3-benzoate (EB, Sigma-Aldrich, catalogue No. E8875) in 100 μL sesame oil at 09:00 hours, two hours after lights on. This protocol evokes an LH surge the following day (Fig.7 [DOI](#)).

To assess the relationship between RP3V^{KISS} neuron and the proestrous LH surge, female mice were attached to the fibre photometry system in the morning of proestrus and tail-tip blood samples (5-μL) collected every 2-4 hours over a period of 8-10 hours (Czieselsky, Prescott et al. 2016 [DOI](#)). Levels of LH were measured by LH ELISA (Kreisman, McCosh et al. 2022 [DOI](#)) with a sensitivity of 0.07 ng/mL and intraassay and interassay coefficients of variation of 8.2% and 11.1%, respectively.

Immunohistochemistry

Female mice receiving bilateral injections of Cre-dependent AAVs encoding GCaMP6s were perfused with ice-cold 4% paraformaldehyde (PFA) on the afternoon of proestrus at least three weeks after viral injections. Brains were removed and post-fixed in 4% PFA for 4 hours before

transferred to 30% sucrose. Forty- μ m-thick coronal sections were cut on a freezing microtome for immunofluorescence. Brain sections were incubated with polyclonal rabbit anti-kisspeptin antisera, raised against the final ten amino acids of murine kisspeptin (1:5,000; AC566, Dr. Alain Caraty, Nouzilly, France) and chicken anti-GFP (1:5,000 Aves Lab, catalogue No. GFP-1020) for 72 hours. Brain sections were washed and incubated with biotinylated goat anti-rabbit IgG (1:500; Vector Laboratories, catalogue No. BA-1000) followed by Streptavidin Alexa Fluor 568 (1:500; Invitrogen, Thermofisher, catalogue No. S11226) and goat anti-chicken Alexa Fluor 488 (1:1,000; Molecular Probes, Thermofisher, catalogue No. A-11039).

Immunofluorescent images were taken on a laser scanning confocal microscope (Zeiss LSM900) using a 20x objective lens in z-stacks with 2 μ m steps across the depth of the slide. The numbers of GFP-expressing, kisspeptin-expressing, and co-expressing cell bodies were manually counted using ImageJ.

GCaMP6 fibre photometry and blood sampling

GCaMP6 fibre photometry was set up as previously described (Clarkson, Han et al. 2017 [↗](#)). Fluorescence signals were sampled at 10 Hz using a scheduled mode (2 seconds on/ 2 seconds off) of light emission with custom software (Tussock Innovation). Recordings were made for 22-hours from mice across each stage of the estrous cycle, 1 week after ovariectomy, and during different times of the OVX+E2+EB surge protocol (Fig.7 [↗](#)); with all starting 4 hours after lights-on.

Analysis of fluorescent signals was performed in MATLAB with the subtraction of the 405 signal from the 465 signal to extract the calcium-dependent fluorescence signal. Fluorescent signals were then down-sampled by averaging each 2-second recording during the 4-second on-off period into a single data point. An exponential fit algorithm was used to correct for baseline shifts using a 6-hour window before the signal was calculated in dF/F (%) with the equation $dF/F = (F_{\text{fluorescence}} - F_{\text{baseline}}) / F_{\text{baseline}} \times 100$.

Numpy.trapezoid function was used in Python to calculate the definite integral of the curve, approximating the area under the curve (AUC) using the trapezoidal rule. Deconvolution of signals was achieved by applying the movmean algorithm in MATLAB to extract a 30-minute rolling average (Han, Yeo et al. 2025 [↗](#)). With this approach, slow oscillations in baseline were then identified by findpeaks function in MATLAB with a) a minimum peak height set as two standard deviations above the mean calcium signal from 11:00-13:00 hours, b) minimum peak distance as 30 minutes, and c) minimum peak prominence as 0.15 of maximum signal in proestrous recording for each animal. Time taken between one trough to another in individual oscillations was calculated as the duration of each oscillation. The total duration of the surge signals was measured from their onset to offset. Onset was defined as the trough at the start of the first oscillation, and offset was defined as the trough at the end of the last oscillation.

To analyse the high frequency transients, the 30-minute moving average was subtracted from the original signal, and an additional exponential fit algorithm with a 400-second window was applied in MATLAB to correct baseline shifts. To identify significant calcium transients, the method of Dombeck and colleagues (Dombeck, Harvey et al. 2010 [↗](#)) was adapted for GCaMP6s dynamics. First, the dF/F trace was z-scored such that its median was normalised to zero, and the standard deviation was set to one. Then, a series of thresholds iterated over the range 1 to 4 with step size 0.2: $k = [1, 1.2, 1.4, \dots, 4]$ were defined. For each threshold k : positive and negative-going transients (T_{pos} and T_{neg}) were identified. The z-scored trace was then thresholded such that $z\text{-score} > k$ to generate a binary array. The segments of the array labelled as 1 (active transient) were classified as T_{pos} . The z-scored trace was also thresholded so that $z\text{-score} < -k$, producing another binary array, with segments labelled as 1 classified as T_{neg} . For each transient in T_{pos} , the transient significance was also assessed for its duration size s . The number of transients with sizes larger than s in both T_{pos} and T_{neg} were counted, as n_{pos} and n_{neg} respectively. If $n_{\text{neg}} / n_{\text{pos}} < 0.05$, the positive transients with sizes greater than s were identified as significant. Any part of the trace identified as significant under any combination of k and s was considered a significant transient.

Statistical Analysis

All statistical analyses were performed using GraphPad Prism 10 software. Each dataset was tested for normality with Shapiro-Wilk test and visual inspection of Q-Q plots. Homogeneity of variance was tested by F-test when comparing two groups or Bartlett's test when comparing three or more groups. If data are normally distributed and variances are homogeneous, one-way ANOVA, repeated measures one-way ANOVA with Geisser-Greenhouse correction followed by Tukey's multiple comparisons test, and paired t-tests were applied. Non-parametric tests, such as Friedman test followed by Dunn's multiple comparison tests were used when data are not normally distributed, or when the variances are not homogeneous. The threshold level for statistical significance was set at $p < 0.05$.

Data availability

All data generated or analyzed during this study are included in the manuscript and supporting files; source data files have been provided for all figures.

Acknowledgements

This work was supported by the Wellcome Trust (212242/Z/18/Z) and a UKRI Medical Research Council Equipment Grant (MC-PC-MR-X012271/1). ZZ was supported by the UKRI Medical Research Council (MR N013433-1) and Harding Distinguished Postgraduate Scholars Programme Leverage Scheme. For the purpose of open access, the author has applied a Creative Commons Attribution (CC BY) license to any Author Accepted Manuscript version arising from this submission.

Additional information

Author Contributions

ZZ undertook investigations; ZZ and CYH performed analysis; ZZ and AEH designed experiments and wrote the manuscript.

Funding

Funder	Grant reference number	Author
Wellcome Trust (WT)	https://doi.org/10.35802/212242	Allan Edward Herbison
UKRI Medical Research Council (MRC)	N013433-1	Ziyue Zhou

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Peer reviews

Joint Public Review:

Summary:

This is an excellent, timely study investigating and characterizing the underlying neural activity that generates the neuroendocrine GnRH and LH surges that are responsible for triggering ovulation. Abundant evidence accumulated over the past 20 years implicated the population of kisspeptin neurons in the hypothalamic RP3V region (also referred to as the POA or AVPV/PeN kisspeptin neurons) as being involved in driving the GnRH surge in response to elevated estradiol (E2), also known as the estrogen positive feedback. However, while former studies used cfos coexpression as a marker of RP3V kisspeptin neuron activation at specific times and found that this correlates with the timing of the LH surge, detailed examination of the live in vivo activity of these neurons before, during, and after the LH surge, remained elusive due to technical challenges. In this exciting study, Zhou and colleagues use fiber photometry to measure the long-term synchronous activity of RP3V kisspeptin neurons across different stages of the mouse estrous cycle, including on proestrus when the LH surge occurs, as well as in a well-established OVX+E2 mouse model of the LH surge. For this they used kiss-Cre female mice that were injected with a Cre-dependent AAV injection containing GCaMP6, in order to measure the neuronal activation of RP3V Kiss1 cells.

The authors report that RP3V kisspeptin neuronal activity is low on estrous and diestrus, but increases on proestrus several hours before the late afternoon LH surge, mirroring prior reports of rising GnRH neuron activity in proestrus female mice. The measured increase in RP3V kisspeptin activation is long, spanning ~13 hours in proestrus females and extending well beyond the end of the LH secretion, and is shown by the authors to be E2 dependent. In addition, an intriguing cyclical oscillation in kisspeptin neural activity every 90 minutes exists, which may offer critical insight into how the RP3V kisspeptin system operates.

The compelling methodology allowed the authors to measure RP3V neuronal activation across multiple ovarian cycles in the same mouse, which demonstrated that the timing of the LH surge is variable across cycles, even within the same mouse. In addition, the authors demonstrated using the same females, that ovariectomy resulted in very little neuronal activity in RP3V kisspeptin neurons. When these ovariectomized females were treated with estradiol benzoate (EB) and an LH surge was induced, there was an increase in RP3V kisspeptin neuronal activation, as was seen during proestrus. However, the magnitude of the change in activity was greater during proestrus than during the EB-induced LH surge. Interestingly, the authors noted a consistent peak in activity about 90 minutes prior to lights out on each day of the ovarian cycle and during EB treatment, but not in ovariectomized females. The functional significance of this consistent neuronal activity at this time remains to be determined. In summary, the data from these experiments is compelling and supports the hypothesis in the field that the RP3V kisspeptin neurons regulate the LH surge.

Strengths:

- The study is well designed, uses proper controls and analyses, has robust data, and the paper is nicely organized and written.
- The study is well done and complete, looking at neuronal activation at each stage of the ovarian cycle and then additionally, how neuronal activation in ovariectomized and ovariectomized + EB females compares to that of gonad-intact females. Though not part of

this study, the comparison of neuronal activation of GnRH neurons during the LH surge to the current data was convincing, demonstrating a similar pattern of increased activation that precedes the LH surge.

- The authors provide a technical advance for the field in the ability to accurately measure RP3V kisspeptin neuron activity in actively awake, live mice for long periods of time, spanning different cycle stages. This approach offers novel and useful insights into the impact of E2 and circadian cues on the electrical activity of RP3V kisspeptin neurons.
- The within-subjects design used in these experiments is a major strength because it allowed the authors to collect data across multiple ovarian cycles, following ovariectomy, and then with EB treatment. The variability in neuronal activity surrounding the LH surge across ovarian cycles in the same animals is interesting and could not be achieved without this within-subjects design.
- The inclusion and comparison of ovary-intact females and OVX+E2 female is valuable to help test mechanisms under these two valuable LH surge conditions, and allows for further future studies to tease apart minor differences in the LH surge pattern between these 2 conditions.
- The discovery of cyclical oscillation in RP3V kisspeptin neural activity every 90 minutes is intriguing and interesting, and may offer critical insight into how the RP3V kisspeptin system operates, which can be further tested in future studies.

Weaknesses:

- LH levels were not measured in many mice or in robust temporal detail, to allow a more detailed comparison between the fine-scale timing of RP3V neuron activation with onset and timing of LH surge dynamics. While the "peak LH" occurred 3.5 hours after the first RP3V kisspeptin neuron oscillation, it is likely that LH values start to increase several hours before the peak LH, closer to when the first RP3V kisspeptin neuron activity first occurs. Therefore, the onset of the LH surge is likely to be closer to the beginning of the RP3V kisspeptin activity, but future studies are needed to study this timing.
- One minor concern is that LH levels were not measured in the ovariectomized females during the expected time of the LH surge. The authors suggest that the lower magnitude of activation during the LH surge in these females, in comparison to proestrus females, may be the result of lower LH levels. It's hard to interpret the difference in magnitude of neuronal activation between EB-treated and proestrus females without knowing LH levels. In addition, it's possible that an LH surge did not occur in all EB-treated females, and thus, having LH levels would confirm the success of the EB treatment.
- The authors nicely show that there is some variation (~2 hours) in the peak of the first oscillation in cycling proestrus females. By contrast, the small sample size for OVX+E2 females did not permit a similar rigorous analysis of temporal variability under such estrogen-controlled conditions, which will need to be studied in future projects.

Comments on revisions:

The authors have revised the manuscript adequately. There are no further recommended edits or revisions.

<https://doi.org/10.7554/eLife.109215.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Joint Public Review:

Weaknesses:

(1) LH levels were not measured in many mice or in robust temporal detail, such as every 30 or 60 min, to allow a more detailed comparison between the fine-scale timing of RP3V neuron activation with onset and timing of LH surge dynamics.

Please see “Recommendations for Authors” below.

(2) The authors report that the peak LH value occurred 3.5 hours after the first RP3V kisspeptin neuron oscillation. However, it is likely, and indeed evident from the 2 example LH patterns shown in Figures 3A-B, that LH values start to increase several hours before the peak LH. This earlier rise in LH levels (“onset” of the surge) occurs much closer in time to the first RP3V kisspeptin neuron oscillatory activation, and as such, the ensuing LH secretion may not be as delayed as the authors suggest.

Please see “Recommendations for Authors” below.

(3) The authors nicely show that there is some variation (~2 hours) in the peak of the first oscillation in proestrus females. Was this same variability present in OVX+E2 females, or was the variability smaller or absent in OVX+E2 versus proestrus? It is possible that the variability in proestrus mice is due to variability in the timing and magnitude of rising E2 levels, which would, in theory, be more tightly controlled and similar among mice in the OVX+E2 model. If so, the OVX+E2 mice may have less variability between mice for the onset of RP3V kisspeptin activity.

Please see “Recommendations for Authors” below.

(4) One concern regarding this study is the lack of data showing the specificity of the AAV and the GCaMP6s signals. There are no data showing that GCaMP6s is limited to the RP3V and is not expressed in other Kiss1 populations in the brain. Given that 2ul of the AAV was injected, which seems like a lot considering it was close to the ventricle, it is important to show that the signal and measured activity are specific to the RP3V region. Though the authors discuss potential reasons for the low co-expression of GCaMP6 and kisspeptin immunoreactivity, it does raise some concern regarding the interpretation of these results. The low co-expression makes it difficult to confirm the Kiss1 cell-specificity of the Cre-dependent AAV injections. In addition, if GFP (GCaMP6s) and kisspeptin protein co-localization is low, it is possible that the activation of these neurons does not coincide with changes in kisspeptin or that these neurons are even expressing Kiss1 or kisspeptin at the time of activation. It is important to remember that the study measures activation of the kisspeptin neuron, and it does not reveal anything specific about the activity of the kisspeptin protein.

Please see “Recommendations for Authors” below.

(5) One additional minor concern is that LH levels were not measured in the ovariectomized females during the expected time of the LH surge. The authors suggest that the lower magnitude of activation during the LH surge in these females, in comparison to proestrus females, may be the result of lower LH levels. It's hard to interpret the difference in magnitude of neuronal activation between EB-treated and proestrus females without knowing LH levels. In addition, it's possible that an LH surge did not occur in all EB-treated females, and thus, having LH levels would confirm the success of the EB treatment.

Please see “Recommendations for Authors” below.

(6) This kisspeptin neuron peak activity is abolished in ovariectomized mice, and estradiol replacement restored this activity, but only partially. Circulating levels of estradiol were not measured in these different setups, but the authors hypothesize that the lack of full restoration may be due to the absence of other ovarian signals, possibly progesterone.

Please see “Recommendations for Authors” below.

(7) Recordings in several mice show inter- and intra-variability in the time of peak onset. It is not shown whether this variability is associated with a similar variability in the timing of the LH surge onset in the recorded mice. The authors hypothesized that this variability indicates a poor involvement of the circadian input. However, no experiments were done to investigate the role of the (vasopressinergic-driven) circadian input on the kisspeptin neuron activation at the light/dark transition. Thus, we suggest that the authors be more tentative about this hypothesis.

Please see “Recommendations for Authors” below.

Recommendations for the authors:

(1) The study measured LH levels over time in just 5 female mice, a small sample size given the variability between mice. Having said that, $n=5$ is an OK starting point but the LH values are only shown for 2 mice, and there are no graphs or presentation of mean LH levels over time for all 5 mice. Figure 3 would greatly benefit from graphing and statistical analyses of the LH levels for all 5 mice (mean line graphs over time or similar). The authors report the mean “peak LH” level in the text, but it would be important to show and graph all the LH values over time (either by clock time or time relative to start of first RP3V oscillation or both), to allow the reader to compare the LH pattern to the RP3V kisspeptin neuron activity over time.

We share the Reviewer’s frustration regarding the lack of detailed LH time points to correlate with the changes in GCaMP signal. Certainly, it was our intention to do better. However, with the benefit of actually being able to monitor surge progress through RP3V neuron activity in real time, we found that frequent blood sampling could often interfere with the normal dynamic of surge activity. On some occasions, the RP3V kisspeptin neuron oscillations would stop abruptly mid- or early-surge while on others it would stop and then start again. Knowing that this was not the normal profile, we resorted to taking as few blood samples as possible, trying primarily to get what we thought might be the “peak” LH surge level. We acknowledge that this is not ideal, and leaves open the important question around the precise relationship of the beginning of RP3V kisspeptin oscillations with LH secretion. Although not answering the question directly, this was part of the motivation for the last figure which emphasizes how the RP3V kisspeptin neuron activity and GnRH neuron dendron activity are essentially identical at the time of the surge. We have re-written the relevant section of the Discussion to be more circumspect.

(2) The authors report and discuss that the peak LH value occurred 3.5 hours after the first RP3V kisspeptin neuron oscillation but it is likely, and indeed evident from the 2 example LH patterns shown in Figs 3A-B, that LH values start to increase several hours earlier, well before the peak LH. Thus, the rise in LH levels during the surge starts much closer in time to the first RP3V kisspeptin neuron oscillatory activation, which the authors don’t analyze. For example, the 2nd LH value for the 2 representative mice shown in Figure 3 is notably higher than the 1st LH value of those mice, even though the peak value has not yet been attained. Even with the LH levels only being measured here every couple hours, this “first detected rise in LH” be at least be graphed and/or analyzed relative to the timing of kisspeptin neuron activity, and commented on in the Discussion.

As above.

(3) It is unclear if the variation (~2 hours) in the peak of the first oscillation in proestrus females is the same as in OVX+E2 females, or was the variability smaller or absent in OVX+E2 females versus proestrus? The variability observed in proestrus mice is likely due to variability in timing and magnitude of rising E2 levels, which would may be more tightly controlled and similar among mice in the OVX+E2 model. If so, the OVX+E2 mice might display less variability for the timing of the RP3V kisspeptin activity "onset". This measure would be important to analyze here and to discuss, given that many labs around the world often use an OVX+E2 model.

This is an interesting point given the dogma surrounding the role of the SCN in initiating the surge. Three of the five OVX+E2 mice exhibited clearly discernible GCaMP oscillations that started at approximately noon, 1pm and 2pm. While this sample is very small, it does suggest that the onset of RP3V kisspeptin neuron activity is variable as found in proestrous mice. We have indicated this cautiously given the sample size.

(4) If looking at kisspeptin immunoreactivity is problematic, is it possible to look at Kiss1 RNA levels or to look at Cre-recombinase protein levels? While the Cre-recombinase would just be a proxy for Kiss1/kisspeptin, it may result in higher expression and better co-localization with the GCaMP6s.

Yes, RNAscope would likely be the ideal method to settle this long running issue of apparently poor Kiss-cre targeting in the RP3V. Unexpectedly, however, we found that the mCherry probe bound to Kiss1 in our attempts at an RNAscope evaluation. The use of Cre as a proxy for identifying kisspeptin neurons would almost certainly generate better co-localization as Cre is being used to target GCaMP.

Minor

(1) It was not clear in the manuscript how many cells were counted or contributed to the neuronal activation data. Is it the entire population of RP3V Kiss1 cells? Just a subset? How much variability is there in the number of cells measured/counted between animals? Presumably, the brains were extracted to confirm the placement of the optic fiber. Were there neuroanatomical studies also done on these animals to confirm how many cells express GFP (GCaMP6) and the correct placement and specificity of the AAV? Is there any potential that cells in the BnST or even the ARC took up the virus and were included in these measurements?

It is very difficult if not impossible to establish just how many RP3V kisspeptin neurons contribute to the GCaMP population signal using fibre photometry. This will depend on levels of AAV transfection, distance from the optic fibre, and the numbers of RP3V kisspeptin neurons actually involved in the surge mechanism. Of note, C-Fos data suggest that only around one-third of RP3V kisspeptin neurons are activated at the time of the surge. All fibre placements were subsequently shown to be running alongside GCaMP-expressing AVPV/anterior periventricular nucleus cells (now noted), but the numbers of transfected cells were not quantified. As shown in Fig.4, the GCaMP signal was very similar across all mice suggesting little variation in the relationship between transduction, fibre placement and distance.

The RP3V region is approximately 4-5 mm from the ARN. We felt that the possibility that an AAV injection in the RP3V would spill over into the ARN was so remote that we did not assess GCaMP expression in ARN kisspeptin neurons. We have previously determined for the ARN that recordable GCaMP fluorescence only occurs if the optic fibre is within 0.5 mm from GCaMP-expressing neurons. Ultimately, proof that we are not recording from ARN kisspeptin neurons comes from the very different activity patterns reported here for RP3V neurons compared to the kisspeptin pulse generator. We did not see any GCaMP expressed in the BNST.

(2) If it is possible to measure LH levels in the EB surge animals, it would be helpful, at least to confirm that they did surge and to support the proposed idea that LH surge levels are lower in that model.

Unfortunately, as acknowledged in the original text we did not take blood samples from these mice so do not have the data. However, as noted, other studies undertaken by us using the same EB surge paradigm show that peak LH levels are much lower compared to proestrus. In retrospect we do agree that this would have been useful and particularly to establish whether each mouse did show a surge as two of the OVX+EB mice failed to show typical surge-associated oscillations. We have noted this in the Discussion.

(3) For Figure 4F, please add a gray shaded box to the graph to denote the "dark" period (lights off), as was done for Figures 2 and 3. This is important because Figure 4F is making the point that there is a consistent 90-minute oscillation event right before lights off, so it would be helpful to denote the period of lights off on the graph.

There was in fact a very light grey shade, but we have now added a grey bar to make the dark period clearer.

(4) The Title of the paper should include the brain region because this is specifically the RP3V (or preoptic area "POA") kisspeptin neurons that are studied, not other kisspeptin cell populations.

We have added "preoptic area" to clarify

(5) The graphs in Figure 3C-D are from different mice and address a different question than the graphs in Figure 3A-B. This was a bit confusing, and it is recommended that the LH + RP3V kisspeptin activity experiment (Figures 3A-B) be its own figure, and the graphs looking at the detailed oscillatory patterns in Figures 3C-D be their own figure, as the latter are addressing a different question and don't have any LH data.

We have split the figure as requested.

(6) The tiny font size of the X and Y axes of Figures 2 and 3 is very small and hard to read. Can this text please be increased in size a little? By comparison, the font size of the X and Y axes of Figure 4 is bigger and more legible.

Changed.

(7) In the methods for fiber photometry, there is a sentence saying "Twenty two-hour recordings were made..." This was confusing, as it read as if there were twenty 2-hour recordings, when in fact it was one 22-hour recording. The authors should reword or use "22-hour" in this sentence.

Changed.

(8) It's a bit hard to see the difference in color between proestrus 1 and proestrus 2 (both blues) in Figure 6, especially when they overlap. It might be helpful to select a different

| color for one of them.

Changed.

(9) Is the virus from Addgene or just the plasmid? Did Addgene insert the plasmid into the virus, or was that done elsewhere? For purposes of replication, it might be helpful to state the plasmid that was used and the virus that was used, and their origins (e.g., if made by Addgene or donated by another investigator). I was not able to find the virus based on the Addgene number in the manuscript and was getting plasmids with different Addgene #s.

Apologies, the numbering was incorrect. We have now amended to 100842-AAV9 that was packaged by Addgene.

<https://doi.org/10.7554/eLife.109215.2.sa0>